

Semester Project: Client/Server System

ECTS

10

Prerequisites

Main purpose

The purpose is to develop and document a client/server system

The purpose of the PBL part of the course is to promote the students' competencies in collaboration, planning and problem analysis.

Knowledge

Apply all knowledge from all 2. semester courses

Skills

Professional skill learning objectives:

- Document the analysis and design of a software system using UML.
- Devise requirements, use case model, activity diagrams, system sequence diagrams, domain model.
- Design and implement (using SQL) a normalized relational database for data-persistence.
- Apply unit and use case testing based on requirements and code.
- Implement a client/server software system in Java according to design.
- Apply Scrum in combination with Unified Process (UP) and relevant tools in the project work.

PBL skill learning objectives:

- Be able to enforce and develop the group contract
- Can work based on the project group's own problem statement.
- Apply covered theories on personal profiles and cross-cultural aspects in the group.
- Apply knowledge about references and source management.
- Apply academic and technical writing style, report structure and rules for plagiarism.
- Communicate the results of the project work and learning process of the project group in a well-structured manner using technical terms in writing, graphically and orally.

Competences

Professional competency learning objectives:

- Carry out a software project with a clear connection between requirements, analysis, design, implementation, test, and documentation.
- Produce a software design which supports group collaboration.
- Apply relevant design patterns and principles, including SOLID, resulting in a maintainable and extendable software system.
- Perform and explain technical choices.

PBL competency learning objectives:

- Describe potential conflicts in the group and suggest solutions.
- Define and reflect on own learning aims from the current and previous semester.

Topics

Analysis, design, test, documentation, project method, Scrum, Unified Process (UP), UML

Cooperation, conflict management, cultural understanding, personal profiles, communication, group work

Teaching methods and study activities

Group work on project tasks supported by process and technical supervision. Online, video og class modules on key aspects.

Continuous evaluation and feedback

Compulsory feedback meetings, and on-going ad hoc meetings

Programme specific product

A product in the form of a model and a program.

Exam

Exam prerequisites:

1. Attendance ($\geq 75\%$) in SWE1

If the exam prerequisite is not met, the student must submit a written assignment in WISEflow to qualify for re-exam.

Type of exam:

The oral exam is held in groups of 4-5 students on the basis of a report.

Joint presentation of 20 minutes followed by a joint examination of 40 minutes + 10 minutes per group member, including voting.

The SEP2 project and exam must demonstrate an understanding of SWE1 skills and competencies and their application in practice.

During the SEP2 exam, specific SWE1 questions will be asked.

Individual grades are given on the basis of an overall assessment.

At the exam, the student is expected to demonstrate ownership of the project by providing detailed explanations and reflections in response to questions posed by the examiner.

The report must have between 8-12 pages (1 normal page = 2400 characters including space), excluding appendices.

One overall grade is given, which applies to both SEP2 and SWE1.

Internal assessment.

Tools allowed:

All.

Re-exams:

Students who have failed a semester project must make a new project.

They must attend an information meeting at the end of the summer exam period.

At this meeting, students will be notified about the process of the SEP re-exam and students will form project groups.

Furthermore, specific deadlines and exam dates are announced at this meeting. The project must be handed in before the deadline.

There will be no guidance in the period up to hand in.

Exam will take place as the ordinary exam.

Grading criteria

Grading based on the Danish 7-point scale.

Responsible

Steffen Vissing Andersen (sva), Mona W. Andersen (mwa) for PBL related topics

Valid from

01-02-2027

Course type

Compulsory for Software Engineering, 2. semester